[University Name]

Department of Computer Science and Engineering

Undergraduate Thesis Proposal

Evaluating Game-State Fusion for Short-Horizon Ball Action Anticipation in Football

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Primary research direction after literature, dataset, novelty, benchmark, and feasibility audit

Abstract

Ball Action Anticipation (BAA) asks a model to predict which ball-related football actions will occur and when they will occur in a short unobserved future interval. Existing BAA work is primarily driven by visual representations, while separate football action-detection and spotting research has shown that explicit game-state information such as player position, velocity, team relation, and multi-player interaction can complement visual evidence when the target action is already observable. This proposal investigates whether that structured information remains useful when the action must be predicted before it occurs.

The proposed study derives a reproducible BAA protocol from SoccerTrack v2, a public dataset containing full-pitch panoramic video, dense game-state reconstruction (GSR), and ball-action spotting (BAS) annotations. The research compares visual-only, game-state-only, simple fusion, flat-relational, and relation-aware fusion models under strict match-level evaluation. The implementation is designed for limited compute: multi-gigabyte GSR JSON files are streamed once into compact state tensors, 4K video is sampled once through a frozen visual encoder, and normal training operates only on compact features. Before full-scale paid compute is used, a 10-minute mini-pilot will validate the entire raw-data-to-prediction pipeline and measure actual resource consumption. The central contribution is therefore a controlled empirical answer to whether explicit player-level game state provides complementary predictive value for short-horizon football action anticipation.

Contents

1. Background and Motivation

Football video understanding has progressed from action spotting and player tracking toward predictive tasks that ask what will happen next. FAANTRA introduced football Ball Action Anticipation on SoccerNet, predicting future ball actions in unobserved frames over short anticipation windows [1]. SoccerNet 2026 subsequently formalized a challenge in which 30 seconds of observed football context are followed by a 5-second future interval containing the events to be anticipated [2].

BAA is attractive for tactical analysis, automated broadcasting, decision-support systems, and intelligent indexing because it operates before an event becomes visible. However, future action prediction is inherently uncertain. Visual evidence alone may not explicitly represent the spatial relationships that make an action likely, such as the distance between attackers and defenders, pressure near the ball, supporting runs, or the collective shape of the two teams.

Separate work demonstrates that explicit football game state can help action understanding. Ochin et al. combine player game state and visual features with a graph-based model for spatio-temporal action detection [3]. Beyond Pixels extends observed-action reasoning with game-state and longer temporal context [4], while FOOTPASS makes multi-agent tactical context central to player-centric spotting [5]. These works establish that structured state is useful prior art. They also prevent this thesis from claiming that player graphs, game-state fusion, or multimodal football understanding are new by themselves.

The unresolved question targeted here is narrower: *does synchronized explicit player-level game state help when the target action is still in an unobserved future interval?*

2. Problem Statement

Given a historical observation window from a football match, the model must predict a variable-size set of ball actions occurring during the following 5 seconds. Each predicted event contains:

1. an action class,
2. a temporal position within the future interval, and
3. a confidence/actionness score.

The task is not next-action classification. More than one action may occur in a 5-second future interval, so the prediction head must support set prediction.

The benchmark will provide up to 30 seconds of historical context for compatibility with the SoccerNet 2026 BAA framing [2]. The initial model will consume a shorter recent context, beginning at 5 seconds, because the original FAANTRA study indicates that simply increasing visual context is not automatically beneficial and increases compute [1]. Context length is therefore a modeling variable rather than a fixed novelty claim.

3. Research Gap and Novelty Boundary

3.1 What prior work already establishes

1. Football Ball Action Anticipation exists and can predict action class and time in unseen future frames [1, 2].
2. Video plus explicit player game state already exists for observed-action football detection [3].
3. Longer temporal and game-level structured context already exists for observed-action denoising/detection [4].
4. Multimodal and multi-agent tactical context already exists in football spotting research [5].
5. Geometry-based football prediction exists in other tasks, including set-piece tactical reasoning and multi-agent tactical forecasting [8, 9, 12].
6. Future football event prediction also exists from sparse event sequences [10, 11].

3.2 Defensible gap

The reviewed BAA literature predicts future actions primarily from visual observations and, in recent challenge work, high-level semantic tactical context. In contrast, explicit positions, velocities, team information, and player-interaction structure have been shown to complement visual evidence for tasks in which the target action is already visible. The literature reviewed for this proposal does not establish whether synchronized explicit player game state provides complementary predictive value for temporally localized ball actions occurring in an unobserved future interval. This thesis tests that question directly.

3.3 Claims deliberately not made

The proposal does not claim to be the first to use a GNN for football players, the first video plus game-state football model, the first football future-event model, or a SoccerNet leaderboard state-of-the-art system. Relation-aware modeling is treated as a controlled method comparison rather than the central invention.

4. Research Questions and Hypotheses

RQ1. Does explicit player-level game-state information improve temporally localized short-horizon BAA compared with visual-only anticipation?

RQ2. Does relation-aware player-interaction modeling provide additional anticipation value beyond simple/flat fusion when the underlying state information is held constant?

H1. Adding explicit game state will improve the mean BAA metric relative to a visual-only baseline.

H2. A relation-aware representation will outperform a flat-relations control that receives the same relational information without message passing.

H3. The effect of game state will differ by action class. This is an exploratory hypothesis. No class-specific improvement is assumed before experiments.

5. Objectives

1. Build a reproducible derived BAA protocol on SoccerTrack v2.
2. Validate BAS, GSR, and video alignment using a pinned dataset revision.
3. Create a compute-efficient preprocessing pipeline for multi-gigabyte GSR JSON and panoramic 4K video.
4. Establish visual-only and game-state-only anticipation baselines.
5. Measure whether simple game-state fusion improves visual anticipation.
6. Measure whether explicit relational features and relation-aware modeling add value beyond flat state.
7. Report grouped match-level performance, fold variation, per-class AP, and efficiency measures.
8. Produce a zero-recurring-cost inference demonstration after the research experiments are stable.

6. Related Work

Table 1: Positioning of the most important related work.

Work	Task	Main input/context	Relevance and remaining difference
FAANTRA [1]	Future BAA	Visual football video	Defines the anticipation task and query-style prediction, but does not use explicit synchronized per-player game state.
SoccerNet 2026 [2]	Future BAA challenge	Mainly visual, plus documented semantic tactical context in one method	Official 30s observation / 5s future framing. Reviewed methods do not document metric GSR/player-graph inputs.
Ochin et al. [3]	Spatio-temporal action detection	Video + positions + velocities + team information + graph	Strongest architectural prior. The target action is observed rather than unobserved future.
Beyond Pixels [4]	Detection/denoising	Player-centric predictions + clean game state + longer context	Weakens novelty of contextual/relational football action modeling, but remains observed-action reasoning.
FOOTPASS [5]	Player-centric spotting	Multimodal, multi-agent tactical context	Further establishes tactical context for spotting, not future BAA.
SoccerTrack v2 [6]	Dataset for GSR/BAS/MOT	Panoramic 4K + player pitch state + BAS	Provides the synchronized data substrate for the proposed derived anticipation protocol.
TacticAI [8]	Corner-kick tactical prediction/generation	Structured player geometry	Shows geometry-based predictive football reasoning is not new.
GenTac [9]	Multi-agent tactical forecasting	Tracking trajectories	Threatens generic tactical forecasting novelty, but not video + GSR BAA.
Seq2Event [10]	Next-event prediction	Sparse event sequences	Shows future football-event prediction itself is not new.
TacticGen [12]	Tactical trajectory generation	Large event + tracking corpus	Multi-agent future generation at much larger scale, but a different prediction target.

7. Research Logic

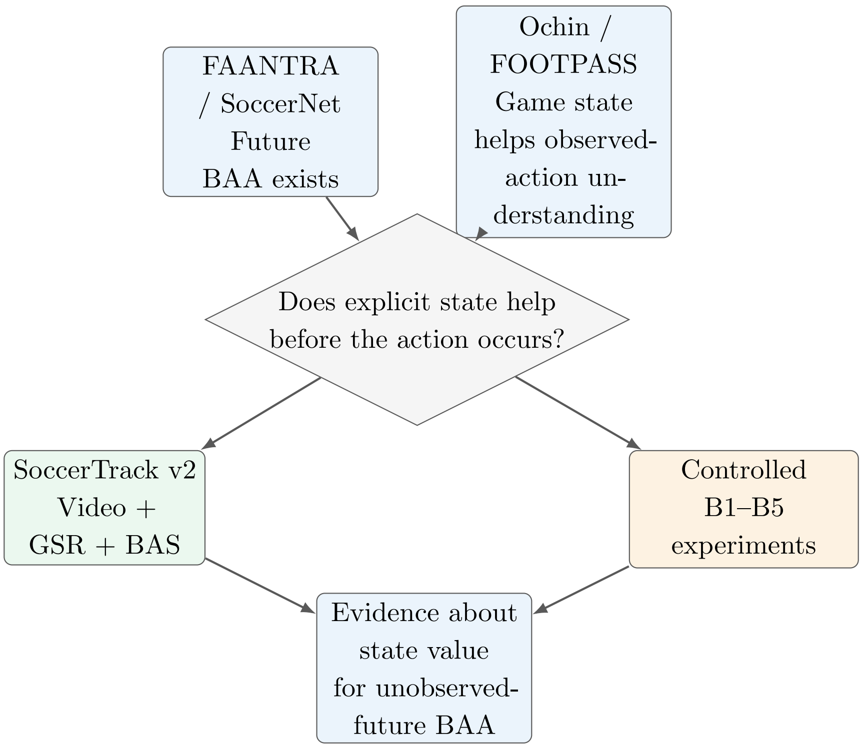

Figure 1: Research-gap logic. The thesis tests a bridge between established visual anticipation and established game-state-aware observed-action understanding.

8. Dataset

8.1 SoccerTrack v2

SoccerTrack v2 is a public dataset containing 10 full-length university-level football matches with panoramic 4K video, dense GSR annotations, and BAS annotations for 12 classes [6, 7]. The GSR includes metric 2D pitch coordinates, player identity/tracking information, roles, and teams. This combination allows the thesis to study future BAA without first training player detection, tracking, re-identification, or camera calibration.

8.2 Historical local BAS audit

A previously downloaded Google Drive snapshot was directly audited during topic selection and contained 23,663 BAS events. Pass and Drive dominated the distribution, while Header, Goal, and Free Kick were rare. This count is retained only as provenance and feasibility evidence. It is **not** treated as the final benchmark count because the Drive-mirror schema differs from the currently documented release structure. Final counts will be regenerated from the pinned experimental revision.

8.3 Known release-quality safeguards

The public repository contains evolving documentation and correction notes [7]. The thesis will therefore:

1. pin the experimental release/revision,
2. record file names and hashes when practical,
3. use project loaders for timestamp semantics,
4. validate actual shipped schema before writing custom parsers,
5. quarantine rather than silently delete suspicious annotations, and
6. resolve or exclude any match with a documented correction that affects the required GSR fields.

Match 132831 is treated as a special audit case because the repository documents a calibration correction issue. Exact benchmark folds are not frozen until this gate passes.

9. Derived BAA Benchmark Protocol

9.1 Task

Each sample contains up to 30 seconds of available historical context followed by a 5-second unobserved future interval. The target is the set of BAS actions occurring in that future interval.

9.2 Class policy

The primary result will use the 10 semantic classes aligned with the established SN-BAA class space by excluding Goal and Free Kick. This is a comparability decision, not a claim that those are the two rarest classes in SoccerTrack v2. A fuller 12-class result will also be reported where statistically meaningful. Rare Header performance will be explicitly visible rather than silently removed after observing model results.

9.3 Split and leakage policy

1. Split by whole match, never by random windows.
2. Prefer five-fold grouped evaluation if enough corrected matches remain usable.
3. No sample crosses half boundaries.
4. No future video, GSR, or BAS information enters the input.
5. Normalization and class weights are computed from training matches only.
6. Player identity is used for trajectory continuity, not as a default semantic learned embedding.
7. Evaluation windows are generated independently of future-event presence.

9.4 Metrics

The evaluation follows BAA-style mean Average Precision at temporal tolerances $\delta = 1, 2, 3, 4, 5$ seconds and $\delta = \infty$ [1, 2]. For finite δ, a prediction is temporally correct when it falls within $\delta/2$ seconds of the ground-truth time. We will report per-class AP, aggregate mAP, fold mean and dispersion, and avoid comparing absolute SoccerTrack results to the SoccerNet leaderboard because the datasets and domains differ.

10. Large-Data System Architecture

The raw dataset is large because dense 25-fps GSR is stored in large JSON files and the video is panoramic 4K. The proposed pipeline therefore converts raw modalities once into compact continuous feature stores.

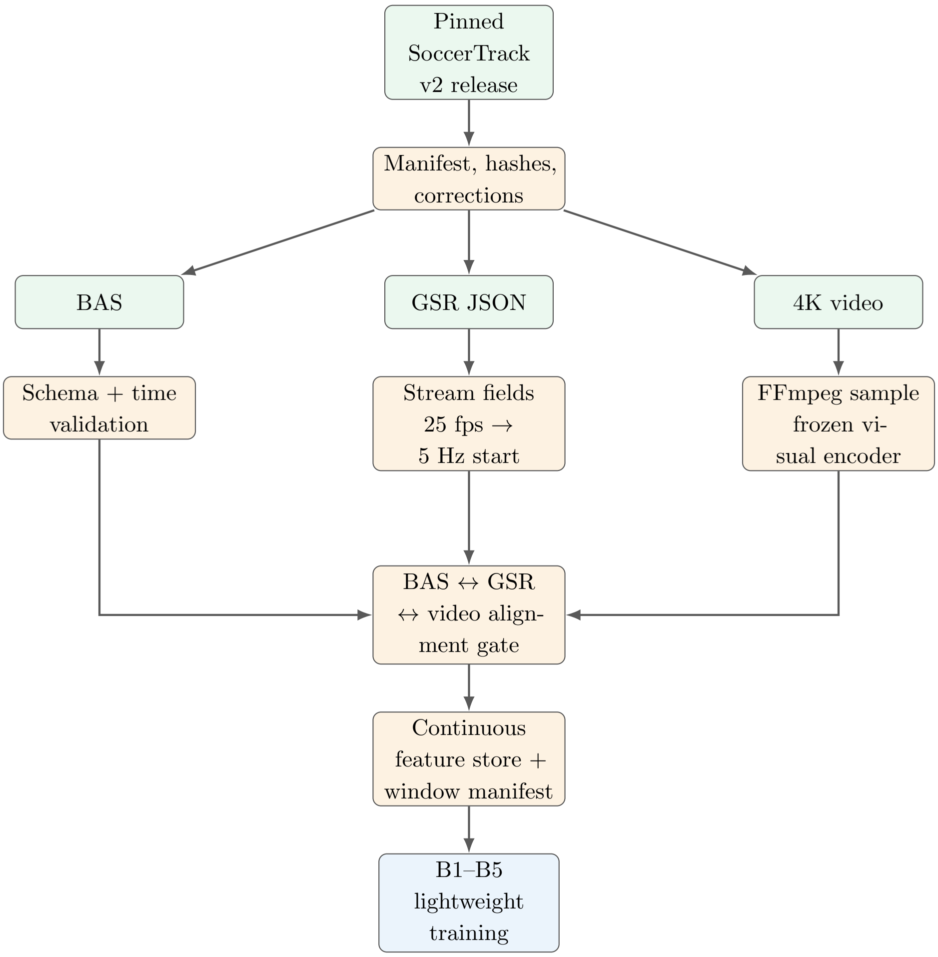

Figure 2: Raw-dataset-to-model architecture. Large raw files are processed once; model training uses compact aligned artifacts.

10.1 GSR streaming and compression

The GSR parser will process one half at a time with an incremental parser or official loader. It will keep only task-relevant information: time/frame, player pitch position, role/team relation, tracking identifiers for continuity, and validity masks. Velocity will be derived from historical positions. The first implementation will sample state at 5 Hz. This is an engineering starting point, not a claim of optimal temporal resolution.

A fixed/padded player tensor plus mask will be produced for batching. Continuous per-half arrays will be saved once. Pairwise relations such as relative position, distance, relative velocity, and same-team/opponent indicators can be computed dynamically, preventing a large permanently materialized edge store.

For approximately 900 minutes of football, 5 Hz yields about 270,000 state timestamps. A dense tensor with 22 players and 8 FP16 values per timestamp is only on the order of 100 MiB before metadata and masks, demonstrating why large JSON can become compact model-ready data.

10.2 Video feature extraction

Video will be copied/downloaded one match or half at a time, sampled at approximately 6.25 fps as a starting configuration, resized, and passed through a frozen pretrained visual encoder. Embeddings are stored in FP16 with timestamps, then the temporary runtime copy of the raw media can be removed. For roughly 900 minutes, 6.25 fps corresponds to about 337,500 sampled timestamps. A 768-dimensional FP16 representation is roughly 0.5 GiB before metadata, making repeated raw-video decoding unnecessary during training.

10.3 Window indexing instead of clip duplication

Overlapping training/evaluation windows will not be saved as separate duplicated tensors. The system stores continuous per-half visual/state arrays and a small table with match, half, context start/end, future start/end, and target-event indexes. The data loader slices these arrays at runtime. This substantially reduces storage and I/O.

11. Cross-Modal Validation and Double-Checking

Every BAS event enters the benchmark only after it passes:

1. accepted schema under the pinned loader,
2. known half/segment,
3. mappable GSR time/frame,
4. corresponding video time,
5. no unresolved correction affecting required state.

Failures are recorded with reason codes in an exclusion ledger. The preprocessing stage will generate a dataset manifest, correction log, alignment report, class distribution, per-match counts, usable durations, and final fold manifest.

Approximately 100 randomly sampled aligned events will also receive a manual quality spot-check using the BAS label/time, short video context, and GSR/minimap state. This uses the small manual-review budget as quality assurance rather than as large-scale new annotation.

12. Proposed Model

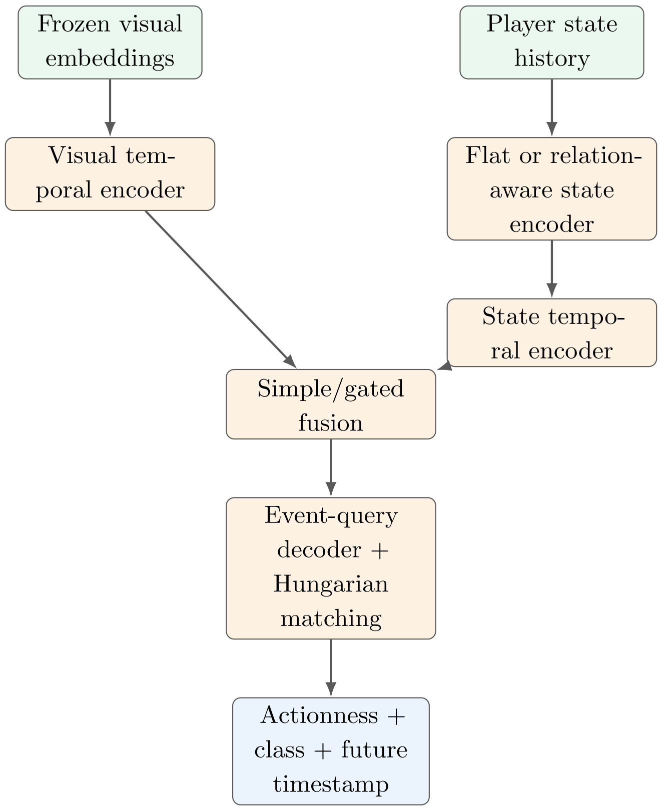

Figure 3: Proposed lightweight dual-branch model family. Complexity is intentionally limited so the scientific value comes from controlled comparisons.

12.1 Visual branch

A frozen visual backbone produces features that are summarized by a small GRU or lightweight temporal encoder. The backbone is not the novelty and will not be repeatedly fine-tuned on 4K frames.

12.2 Game-state branch

Player nodes include normalized x, y, derived v_x, v_y, role, masks, and team relation. Relation-aware variants use local player interactions with features such as Δx, Δy, distance, relative velocity, and same-team/opponent indicators. A small message-passing or attention encoder is sufficient. The first design targets two relational layers rather than a deep graph network.

12.3 Set-prediction head

A fixed number of event queries predicts actionness, class, and normalized future timestamp. Predictions are matched to variable-size ground-truth event sets using Hungarian matching. Classification can use class weighting; timestamp uses an L1-style regression term; unmatched queries represent no-event slots.

13. Experiment Matrix

Table 2: Core controlled experiment ladder.

ID	Model	Scientific purpose
B0	Statistical/no-event floor	Sanity floor for the derived benchmark.
B1	Visual-only BAA	Establish the visual anticipation baseline.
B2	Game-state-only	Test whether explicit state alone contains anticipatory information.
B3	Simple fusion	Test RQ1: whether adding state improves visual-only anticipation.
B4	Flat-relations fusion	Give explicit relational features without graph message passing.
B5	Relation-aware fusion	Test RQ2 by comparing against B4 with equivalent information.

The critical interpretation is B3 versus B1 for the main research question and B5 versus B4 for the relation-model question. If B3 improves while B5 does not beat B4, the thesis can conclude that state is useful without requiring graph reasoning. If B5 fails to beat B1, the result can still show that game-state value observed in detection does not automatically transfer to anticipation.

13.1 Ablations

Mandatory comparisons are no-game-state, no-visual-input, and relation-aware versus flat-relations. Velocity and team-relation removals are secondary ablations if time permits. The project explicitly rejects an architecture zoo involving many backbones, graph operators, fusion blocks, frame rates, or context lengths before the core research questions are answered.

14. Mini Feasibility Prototype

Before full paid experimentation, a short pilot will prove the entire pipeline on approximately the first 10 minutes of a relatively clean match/half, preferably match 117093 after confirming the pinned release.

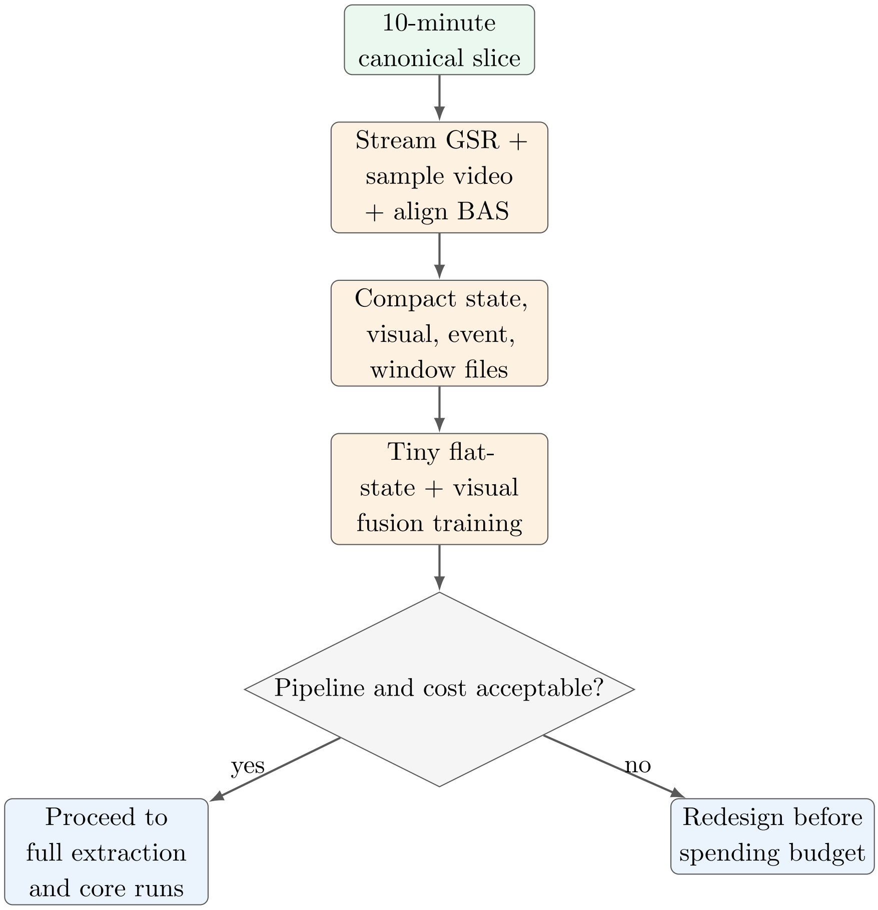

Figure 4: Feasibility pilot as a go/no-go gate before full paid compute.

The pilot must demonstrate successful download/access, streamed GSR parsing, BAS/GSR/video alignment, compact storage, frozen feature extraction, training with finite/decreasing loss, checkpoint save/load, action-class output, timestamp output, and evaluator execution. Accuracy is not a pilot criterion.

The pilot will save a manifest, state tensor, visual features, event/window tables, checkpoint, training log, resource-usage log, and sample predictions. Runtime, peak RAM/VRAM, output size, and accelerator usage are recorded and conservatively extrapolated.

15. Compute Strategy

Google Colab provides dynamically allocated GPU/TPU resources and does not guarantee a fixed accelerator type or exact usage limit [13]. Therefore the proposal treats the anticipated 100-unit paid budget as a soft envelope rather than a promise of exact GPU-hours.

Table 3: Soft accelerator budget. CPU-only work should not consume paid accelerator units where avoidable.

Stage	**Soft ceiling**
Pilot visual extraction	5 units
Remaining full visual extraction	15 units
Pipeline/model debugging	10 units
B1–B5 core experiments	40 units
Essential ablations	15 units
Emergency/re-run reserve	15 units
Total envelope	100 units

15.1 Hard stop rules

1. Measure 10-minute extraction cost before full extraction.
2. Measure one full match before processing the entire dataset.
3. Do not launch grouped full evaluation until B1–B5 work on one fold.
4. Do not spend accelerator units parsing JSON, calculating statistics, or building fold manifests.
5. If full extraction projects beyond budget, lower frame rate, input resolution, or backbone size before continuing.
6. Optional ablations begin only after the core research question has an answer.

TPU may be used for a fixed-shape pilot if necessary, but GPU is preferred for the final graph-based pipeline because it reduces framework and dynamic-shape complexity. The system itself is not dependent on TPU availability.

16. Validation and Statistical Analysis

Core B1–B5 models should be evaluated across match-grouped folds after the dataset gate is complete. The final report will include:

1. mean and dispersion across folds,
2. per-class AP,
3. paired fold differences for B3-B1 and B5-B4,
4. rare-class discussion,
5. parameter count,
6. training time per fold,
7. peak memory,
8. feature-store size, and

9. inference latency.

The dataset contains only a small number of independent matches, so statistical claims will remain conservative. The thesis will not present cross-validation as creating more independent matches than actually exist.

17. Deployment and System Demonstration

Deployment is a supporting engineering contribution. A short clip or pre-prepared sample is sent to a lightweight FastAPI service, temporarily preprocessed, passed through the anticipation model, and returned as structured predictions such as `future_offset`, `action`, and `confidence`. Raw uploaded media is deleted after processing. Only compact structured output and model/experiment metadata need to be stored. The target is zero recurring paid infrastructure.

18. Risk Register and Mitigation

Risk	**Why it matters**	**Mitigation**
Dataset schema drift	Different mirrors/revisions may not match documentation.	Pin revision, record hashes, validate shipped schema, use official loaders.
Known match correction	A calibration issue can contaminate GSR.	Quarantine affected match until required fields are verified or corrected; otherwise exclude it.
Only 10 matches	Single split may be unstable.	Whole-match grouped evaluation and fold dispersion; no window-level random split.
Class imbalance	Pass/Drive dominate and rare classes have unstable AP.	Pre-register class policy, class-aware losses, per-class AP, transparent rare-class analysis.
Incremental novelty	Video + game-state graphs already exist for detection.	Make future BAA state value the primary research question; B4/B5 control isolates relational modeling.
Visual branch adds little	Full-pitch state may already contain much predictive information.	B1/B2/B3 explicitly measure modality contribution. A strong state-only result is scientifically useful.
Graph adds little	Relations may help without GNN message passing.	Flat-relations B4 is mandatory before B5.
Compute overrun	4K video and huge JSON can waste budget.	One-time feature extraction, streaming compression, mini pilot, stop rules, reserve units.
Domain limitation	University-level full-pitch footage differs from professional broadcast.	Bound conclusions to SoccerTrack v2; no professional-domain generalization claim.

Negative result	Fusion may not outperform visual-only.	The design remains valuable because it tests whether detection-time state benefits transfer to anticipation.

19. Reproducibility Plan

The project will version code in Git and keep raw datasets out of the repository. The release will include configuration files, the dataset manifest, exclusions/corrections, feature-building scripts, benchmark window construction, fold manifests, model configs, training seeds, evaluation scripts, and structured experiment logs. The Obsidian research vault separately preserves decision history and evidence corrections.

The model-ready directory is expected to follow:

```
processed/
  manifest/
  events/<match>.parquet
  state/<match>_<half>.npz
  visual/<match>_<half>.npy
  windows/folds.json
  windows/windows.parquet
  audit/alignment_report.json
  audit/class_distribution.json
```

20. Expected Contributions

1. **Primary scientific contribution:** controlled evidence on whether explicit player-level game state improves short-horizon future BAA.
2. **Secondary scientific contribution:** evidence on whether relation-aware player interaction modeling adds value beyond flat state and flat relational features.
3. **Reproducibility contribution:** a documented procedure for deriving BAA samples from synchronized SoccerTrack v2 video, GSR, and BAS while handling release/version and alignment issues.
4. **Engineering contribution:** a compute-efficient raw-data-to-feature-store pipeline and a zero-recurring-cost inference demonstration.

21. One-Month Execution Plan

Table 5: Compressed execution plan.

Week	Primary objective	Outputs
1	Data gate + feasibility	Pin release, audit schemas/corrections, 10-minute pilot, cost projection, finalize usable matches/folds.
2	Baselines	Full compact features, B0–B3 on one fold, fix evaluator and losses, then grouped B1–B3 runs.
3	Relation study	B4/B5, essential ablations, per-class analysis, efficiency logging.
4	Writing + demo	Final tables/figures, limitations, FastAPI/demo, thesis proposal/paper expansion, reproducibility release.

22. Success Criteria

The thesis is considered scientifically complete if the following are achieved:

1. a validated, reproducible SoccerTrack-derived BAA protocol,
2. at least B1, B2, B3, B4, and B5 evaluated under the same grouped protocol,
3. clear evidence for or against RQ1,
4. a controlled result for RQ2,
5. transparent per-class and fold-level analysis,
6. documented resource usage and dataset-quality decisions, and
7. a reproducible code/data-manifest release and small inference demonstration.

A positive accuracy gain is not itself required for thesis validity. The central requirement is an interpretable answer to the research question under a defensible protocol.

23. Conclusion

This proposal deliberately narrows novelty to a question that survives the literature and dataset audit. Football BAA already exists; explicit game-state fusion and player-relationship modeling already exist for observed-action tasks; and future football-event prediction exists in other representations. The proposed thesis therefore does not attempt to claim those components as first-time inventions. Instead, it asks whether synchronized explicit player game state provides useful predictive information when the target ball action has not yet occurred. SoccerTrack v2 makes this question experimentally accessible, while streaming structured preprocessing, one-time frozen video feature extraction, controlled B1–B5 baselines, and a pilot-before-paid-compute workflow make the project realistic for an undergraduate team with limited GPU resources.

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A. Pre-Experiment Gate Checklist

1. Pinned SoccerTrack v2 revision recorded.
2. BAS/GSR/video file inventory and hashes recorded where practical.
3. Actual shipped schemas inspected.
4. Known correction status checked, especially match 132831.
5. Official loader timestamp semantics verified.
6. 10-minute pilot passes alignment and training smoke test.
7. Actual runtime/storage/accelerator cost measured.
8. Exact class counts regenerated from the pinned revision.
9. Exact usable match set finalized.
10. Exact fold manifest frozen before model training.
11. Number of future-event queries selected from training folds only.
12. Evaluation implementation checked against official BAA metric semantics.

B. Pilot Artifact Checklist

```
pilot/
  pilot_manifest.json
  pilot_state.npz
  pilot_visual.npy
  pilot_events.parquet
  pilot_windows.parquet
  model_checkpoint.pt
  training_log.csv
  resource_usage.json
  sample_predictions.json
```